

SPIKE GATE PHASE 2 RESULTS

Audit comparison and 182-galaxy MTOjects diagnostic batch

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EXECUTIVE CONCLUSION

Phase 2 removed the galaxy-scale overmasking failure mode. MTOjects is the stronger diagnostic candidate; SEP failed the stress-test audit and should not enter a production-style batch in its present configuration.

1. STRESS-TEST AUDIT

SEP: objective 153.6543 (infeasible); mean gate recovery 11.1%; candidate detection 11.7%; zero detections 11/15; mean masked area 0.00258%. Decision: do not proceed.

MTOjects: objective 57.8821 (narrowly infeasible); mean gate recovery 49.3%; candidate detection 63.3%; supported precision 58.1%; zero detections 2/15; mean masked area 0.00500%. Decision: proceed diagnostically.

The two MTO zero-detection cases were NGC4020 and NGC4532.

2. NGC3627

SEP produced zero retained components and zero recovery. MTO produced three gate-supported components, recovery 1.0 and candidate detection 1.0. MTO masked 0.0146% while preserving spiral structure and isophotal alignment.

3. 182-GALAXY MTO DIAGNOSTIC BATCH

Successful: 182/182; failed: 0.

Gate-positive: 127. Gate-negative: 55; all left unmasked.

Accepted non-zero mask: 97/127 gate-positive galaxies (76.4%).

No accepted component: 30/127 gate-positive galaxies (23.6%).

Overall: 97 non-zero masks and 85 zero masks.

4. MASKED-AREA DISTRIBUTION

Mean 0.002778%; median 0.000685%; P75 0.003443%; P90 0.010300%; P95 0.013278%; P99 0.016678%; maximum 0.021179% (NGC6412).

5. HIGH-MASK REVIEW

NGC6412 (0.02118%): two compact off-centre masks; galaxy preserved.

NGC1672 (0.01946%): four narrow profile-associated masks; stable isophotes.

NGC3627 (0.01458%): major improvement over the earlier aggressive SEP result.

NGC5236 (0.01428%): 25 accepted from 11,537 raw segments; manual/performance flag.

6. SCIENTIFIC INTERPRETATION

Phase 2 sharply reduces damage to coherent galaxy morphology. MTO is presently the stronger Spike Gate profile-protection candidate. Thirty gate-positive galaxies received no accepted mask, so recall remains incomplete. SEP remains excluded until it reliably yields compact gate-associated components. This remains a conservative diagnostic, not a complete foreground catalogue.

Output: D:\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground
objects\MTO\Spike Gate\20260819_202509